2.51 Graph Sketching: Polar Curves – Part 2

Worked Solutions

1 a) $r = 2\sec\theta$ gives

$$r = \frac{2}{\cos\theta} \quad \Longrightarrow \quad r\cos\theta = 2.$$

1 b) $r = -6\operatorname{cosec}\theta$ gives

$$r = \frac{-6}{\sin\theta} \quad \Longrightarrow \quad r\sin\theta = -6.$$

2 a) From $r = 2\sec(\frac{\pi}{4} - \theta)$,

$$\begin{aligned} r\cos(\tfrac{\pi}{4} - \theta) &= 2, \\ r\cos\tfrac{\pi}{4}\cos\theta + r\sin\tfrac{\pi}{4}\sin\theta &= 2, \\ \tfrac{\sqrt{2}}{2}x + \tfrac{\sqrt{2}}{2}y &= 2, \\ y &= -x + 2\sqrt{2}. \end{aligned}$$

2 b) From $r = 2\operatorname{cosec}(\frac{\pi}{4} - \theta)$,

$$\begin{aligned} r\sin(\tfrac{\pi}{4} - \theta) &= 2, \\ r\sin\tfrac{\pi}{4}\cos\theta - r\cos\tfrac{\pi}{4}\sin\theta &= 2, \\ \tfrac{\sqrt{2}}{2}x - \tfrac{\sqrt{2}}{2}y &= 2, \\ y &= x - 2\sqrt{2}. \end{aligned}$$

2 c) From $r = -3\operatorname{cosec}(\frac{\pi}{6} - \theta)$,

$$\begin{aligned} r\sin(\tfrac{\pi}{6} - \theta) &= -3, \\ \tfrac{1}{2}x - \tfrac{\sqrt{3}}{2}y &= -3, \\ y &= \tfrac{\sqrt{3}}{3}x + 2\sqrt{3}. \end{aligned}$$

2 d) From $r = -4\sec(\frac{5\pi}{6} - \theta)$,

$$\begin{aligned} r\cos(\tfrac{5\pi}{6} - \theta) &= -4, \\ -\tfrac{\sqrt{3}}{2}x + \tfrac{1}{2}y &= -4, \\ y &= \sqrt{3}x - 8. \end{aligned}$$

3 a) The line has gradient $m = -6/(6/\sqrt{3}) = -\sqrt{3}$ and equation $y = -\sqrt{3}x - 6$. Since $x = r\cos\theta$ and $y = r\sin\theta$,

$$r(\sin\theta + \sqrt{3}\cos\theta) = -6.$$

The coefficient amplitude is $\sqrt{1^2 + (\sqrt{3})^2} = 2$, with phase $\pi/6$, so

$$\sin\theta + \sqrt{3}\cos\theta = 2\cos(\theta - \tfrac{\pi}{6}).$$

Therefore $2r\cos(\theta - \frac{\pi}{6}) = -6$, and, since cosine is even,

$$r\cos(\tfrac{\pi}{6} - \theta) = -3.$$

Thus $a = \frac{\pi}{6}$ and $p = -3$.

3 b) The line has gradient $m = -8/(8\sqrt{3}) = -\sqrt{3}/3$ and equation $y = -\frac{\sqrt{3}}{3}x + 8\sqrt{3}$. Hence

$$r(\sin\theta + \tfrac{\sqrt{3}}{3}\cos\theta) = 8\sqrt{3}.$$

The coefficient amplitude is $2\sqrt{3}/3$, with phase $\pi/3$, so

$$\sin\theta + \tfrac{\sqrt{3}}{3}\cos\theta = \tfrac{2\sqrt{3}}{3}\cos(\theta - \tfrac{\pi}{3}).$$

Therefore $2r\cos(\frac{\pi}{3} - \theta) = 8$, giving

$$a = \tfrac{\pi}{3}, \qquad p = 4.$$

3 c) The line has gradient 1 and equation $y = x + 2\sqrt{2}$. Hence

$$r(\sin\theta - \cos\theta) = 2\sqrt{2}.$$

The amplitude is $\sqrt{2}$. Taking the phase in the interval $0 \le \theta \le 2\pi$ gives

$$\sin\theta - \cos\theta = \sqrt{2}\cos(\theta - \tfrac{3\pi}{4}).$$

Therefore $r\cos(\frac{3\pi}{4} - \theta) = 2$, and

$$a = \tfrac{3\pi}{4}, \qquad p = 2.$$

4 a) The line has gradient 1 and equation $y = x - 4\sqrt{2}$. Thus

$$r(\sin\theta - \cos\theta) = -4\sqrt{2}.$$

Since $\sin\theta - \cos\theta = \sqrt{2}\sin(\theta - \frac{\pi}{4})$,

$$\sqrt{2}r\sin(\theta - \tfrac{\pi}{4}) = -4\sqrt{2}.$$

Using the oddness of sine,

$$r\sin(\tfrac{\pi}{4} - \theta) = 4,$$

so $a = \frac{\pi}{4}$ and $p = 4$.

4 b) The line has gradient $-\sqrt{3}/3$ and equation $y = -\frac{\sqrt{3}}{3}x + 2\sqrt{3}$. Hence

$$r(\sin\theta + \tfrac{\sqrt{3}}{3}\cos\theta) = 2\sqrt{3}.$$

The coefficient amplitude is $2\sqrt{3}/3$, with phase $-\pi/6$, so

$$\sin\theta + \tfrac{\sqrt{3}}{3}\cos\theta = \tfrac{2\sqrt{3}}{3}\sin(\theta + \tfrac{\pi}{6}).$$

Since $\sin(\theta + \frac{\pi}{6}) = \sin(\frac{5\pi}{6} - \theta)$,

$$r\sin(\tfrac{5\pi}{6} - \theta) = 3.$$

Thus $a = \frac{5\pi}{6}$ and $p = 3$.

5 a) Expanding the compound angle,

$$\begin{aligned} r\cos(\alpha - \theta) &= k, \\ r\cos\alpha\cos\theta + r\sin\alpha\sin\theta &= k, \\ x\cos\alpha + y\sin\alpha &= k, \\ y &= -\frac{\cos\alpha}{\sin\alpha}x + \frac{k}{\sin\alpha}. \end{aligned}$$

5 b) The printed working gives

$$\begin{aligned} \sin(\alpha - \theta) &= k, \\ r\sin\alpha\cos\theta - r\cos\alpha\sin\theta &= k, \\ x\sin\alpha - y\cos\alpha &= k. \end{aligned}$$

6 a) Since $r^2 = 16\cos 2\theta$, consider

$$r = \pm 4\sqrt{\cos 2\theta}.$$

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6 c) Since $r^2 = 16\cos 2\theta$, consider

$$r = \pm 4\sqrt{\cos 2\theta}.$$

7 a) Since $r^2 = 25\sin 2\theta$, consider $r = \pm 5\sqrt{\sin 2\theta}$.

7 b) Since $r^2 = 10\cos 3\theta$, consider $r = \pm\sqrt{10\cos 3\theta}$.

7 c) Since $r^2 = 12\sin 4\theta$, consider $r = \pm\sqrt{12\sin 4\theta}$.

7 d) Since $r^2 = 9\sin 6\theta$, consider $r = \pm 3\sqrt{\sin 6\theta}$.

7 e) Since $r^2 = 14\cos 5\theta$, consider $r = \pm\sqrt{14\cos 5\theta}$.

7 f) Since $r^2 = 14\cos 5\theta$, consider $r = \pm\sqrt{14\cos 5\theta}$.

8 a) The curve is $r^2 = 16\cos 4\theta$.

8 b) The curve is $r^2 = 9\sin 3\theta$.

8 c) The curve is $r^2 = 36\cos 5\theta$.

9 a) P occurs at the first maximum of $y = \sqrt{64\sin 3\theta} = 8\sqrt{\sin 3\theta}$. So when $\theta = \frac{\pi}{2} \div 3 = \frac{\pi}{6}$. Then

$$x = 8\cos\tfrac{\pi}{6} = 4\sqrt{3}, \qquad y = 8\sin\tfrac{\pi}{6} = 4,$$

so $P = (4\sqrt{3}, 4)$.

9 b) Q occurs at the second maximum of $y = \sqrt{64\sin 3\theta} = 8\sqrt{\sin 3\theta}$. So when $\theta = \frac{\pi}{6} + \frac{2\pi}{3} = \frac{5\pi}{6}$. Then

$$x = 8\cos\tfrac{5\pi}{6} = -4\sqrt{3}, \qquad y = 8\sin\tfrac{5\pi}{6} = 4,$$

so $Q = (-4\sqrt{3}, 4)$.

9 c) At an intersection, $r^2 = 32 = 64\sin 3\theta$, so $\sin 3\theta = \frac{1}{2}$. For $-\pi < \theta \leq \pi$,

$$\theta \in \left\{ \tfrac{\pi}{18}, \tfrac{5\pi}{18}, \tfrac{13\pi}{18}, \tfrac{17\pi}{18}, \tfrac{25\pi}{18}, \tfrac{29\pi}{18} \right\}, \qquad r = \sqrt{32}.$$

The Cartesian coordinates, to 2 d.p., are

$$(5.57, 0.98),\ (3.64, 4.33),\ (-3.64, 4.33),\ (-5.57, 0.98),\ (-1.93, -5.32),\ (1.93, -5.32).$$

10 a) Consider $y = \sqrt{16\cos 4\theta} = 4\sqrt{\cos 4\theta}$. We want values of θ for which $\cos 4\theta > 0$. The printed intervals are

$$\theta \in (0, \tfrac{\pi}{8}) \cap (\tfrac{3\pi}{8}, \tfrac{5\pi}{8}) \cap (\tfrac{7\pi}{8}, \pi).$$

10 b) Similarities: both curves pass through the axes at the same point; the maximum value of r is 4; both curves are symmetric about the axes. Differences: C_1 has four petals while C_2 has eight; C_2 has loops created by negative values of r; the petals of C_1 are wider than those of C_2; and C_2 is continuous while C_1 is discontinuous.